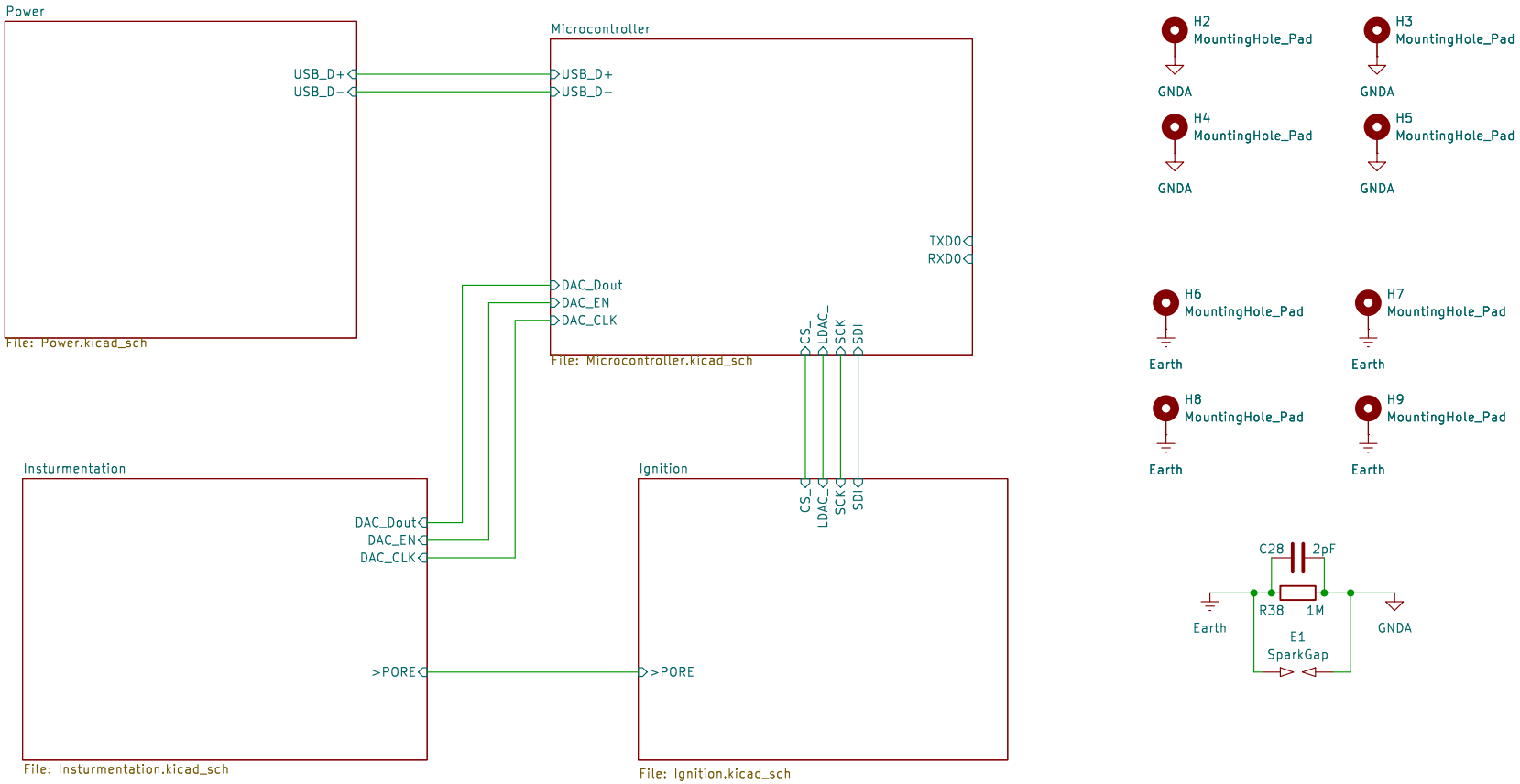
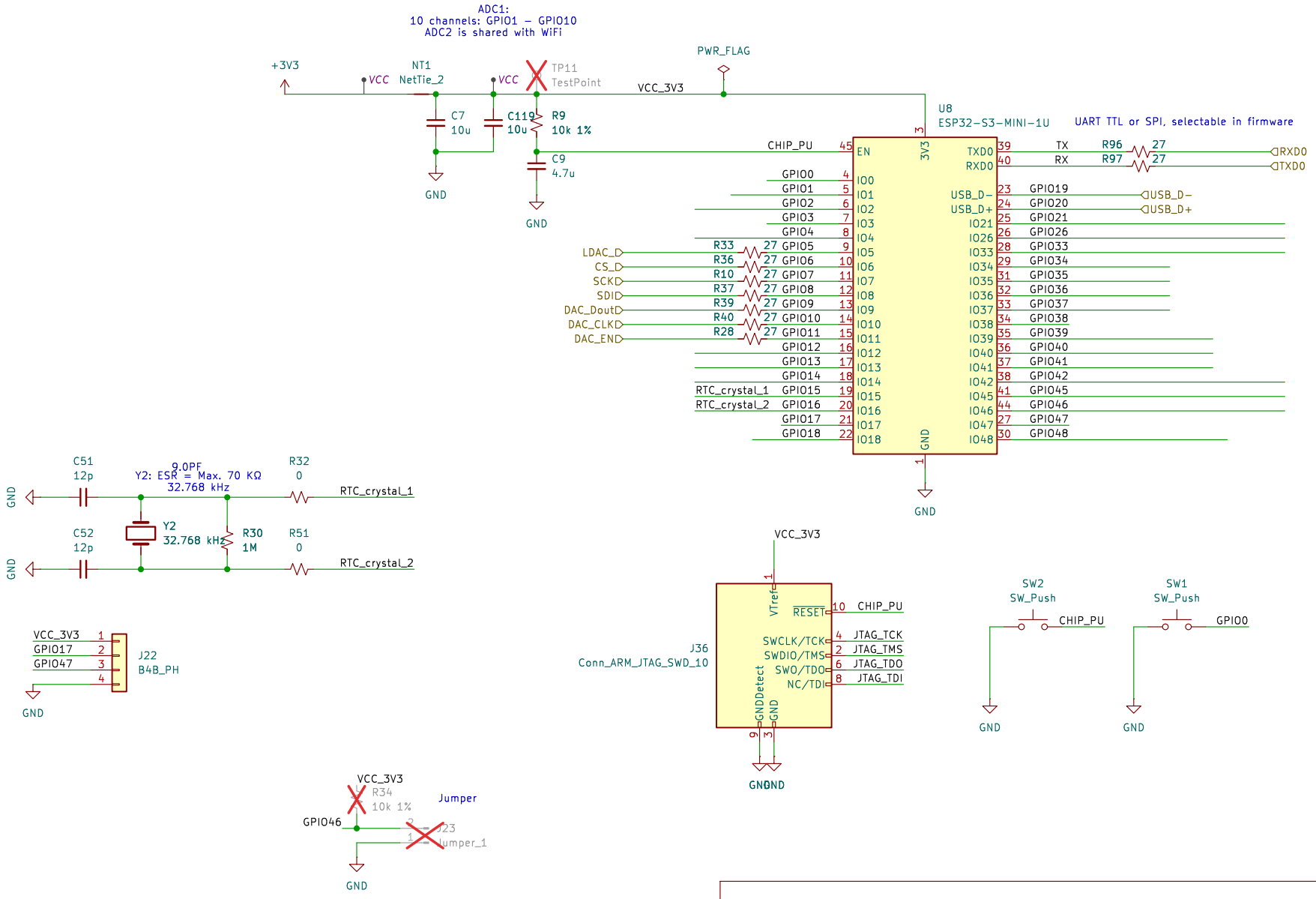
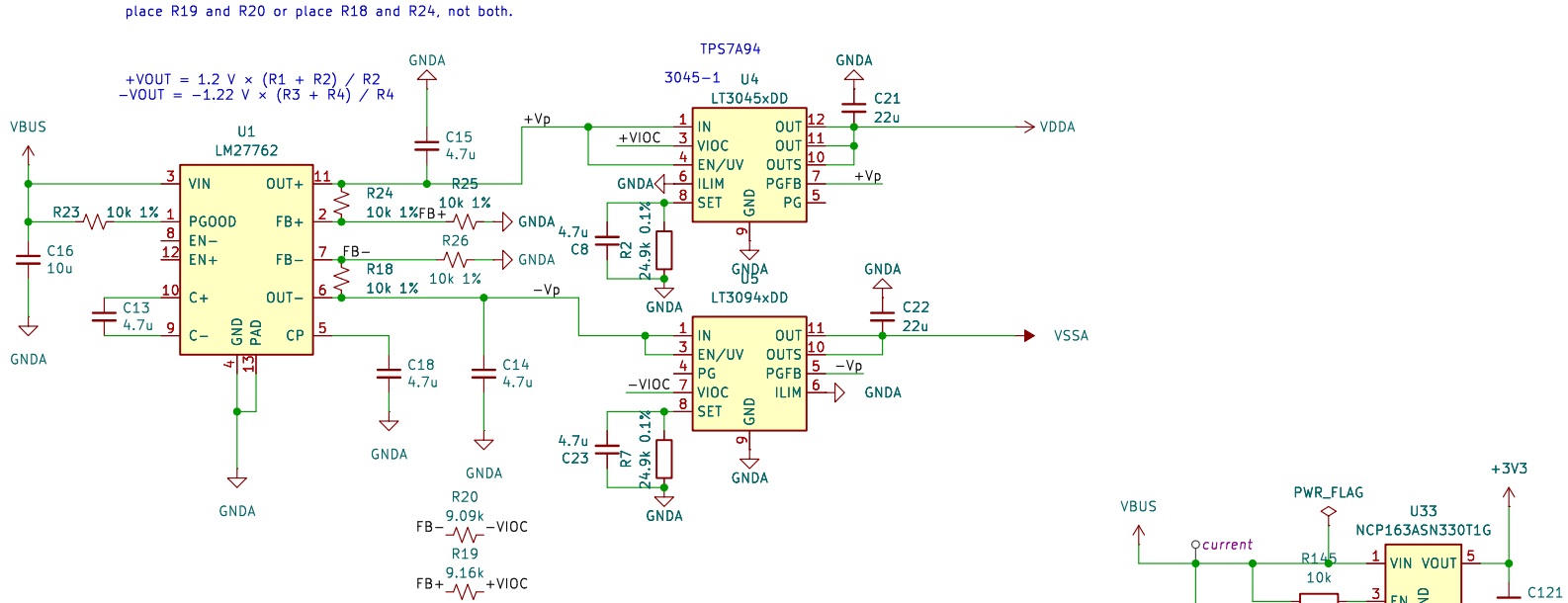


TODO:
update ADC with correct library part and footprint
add voltage reference,

Add precision low noise ADC
Look at Insturmentation amplifiers

Sheet: / File: MR1.kicad_sch		
Title:		
Size: USLetter	Date:	**Rev:**
KiCad E.D.A. 9.0.6		Id: 1/5


ADC1:
10 channels: GPIO1 – GPIO10
ADC2 is shared with WiFi
+3V3
NT1
NetTie_2
VCC
TP11
TestPoint
PWR_FLAG
VCC_3V3
C7
10u
C119
10u
R9
10k 1%
C9
4.7u
GND
U8
ESP32-S3-MINI-1U
UART TTL or SPI, selectable in firmware
CHIP_PU
EN
3V3
TXD0
RXD0
TX
RX
R96
R97
27
RXD0
TXD0
GPIO0
GPIO1
GPIO2
GPIO3
GPIO4
GPIO5
GPIO6
GPIO7
GPIO8
GPIO9
GPIO10
GPIO11
GPIO12
GPIO13
GPIO14
GPIO15
GPIO16
GPIO17
GPIO18
GPIO19
GPIO20
GPIO21
GPIO26
GPIO33
GPIO34
GPIO35
GPIO36
GPIO37
GPIO38
GPIO39
GPIO40
GPIO41
GPIO42
GPIO45
GPIO46
GPIO47
GPIO48
USB_D-
USB_D+
LDAC_
CS_
SCK
SDI
DAC_Dout
DAC_CLK
DAC_EN
R33
R36
R10
R37
R39
R40
R28
RTC_crystal_1
RTC_crystal_2
C51
12p
C52
12p
9.0PF
Y2: ESR = Max. 70 KΩ
32.768 kHz
Y2
32.768 kHz
R30
1M
R32
0
R51
0
J22
B4B_PH
J36
Conn_ARM_JTAG_SWD_10
VTref
RESET
SWCLK/TCK
SWDIO/TMS
SWO/TDO
NC/TDI
GNDDetect
JTAG_TCK
JTAG_TMS
JTAG_TDO
JTAG_TDI
SW2
SW_Push
SW1
SW_Push
R34
10k 1%
Jumper
Jumper_1
Boot Mode | GPIO0 | GPIO46
Default Config | 1(pull-up) | 0 (Pull-down)
SPI Boot | 1 | Any Value
Joint DL Boot | 0 | 0
Invalid | 0 | 1
Sheet: /Microcontroller/
File: Microcontroller.kicad_sch
Title:
Size: A
Date:
Rev:
KiCad E.D.A. 9.0.6
Id: 2/5



Sheet: /Power/	
File: Power.kicad_sch	
Title:	
Size: USLetter	Date:
KiCad E.D.A. 9.0.6	Id: 3/5

amplifier board

MG Chemicals 422C conformal coating
PTFE (ZYF255DA): Dk .approx 2.55

LTC6268, alternate to TI's opamp

low noise voltage converters

Stage 1: Transimpedance Amp LTC6268

499Ω Metal Film Resistor

C2 and C3 X5R

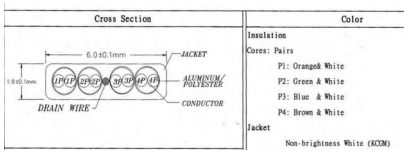
Virtual Ground

consider cat 7 flat ribbon cable, 8 pairs, shielded

pins for holding the board in place while applying adhesive

https://www.mcmaster.com/98378A208/





https://www.vpi.us/bulk-cable-sf/super-flat-cat7-736#idTab9

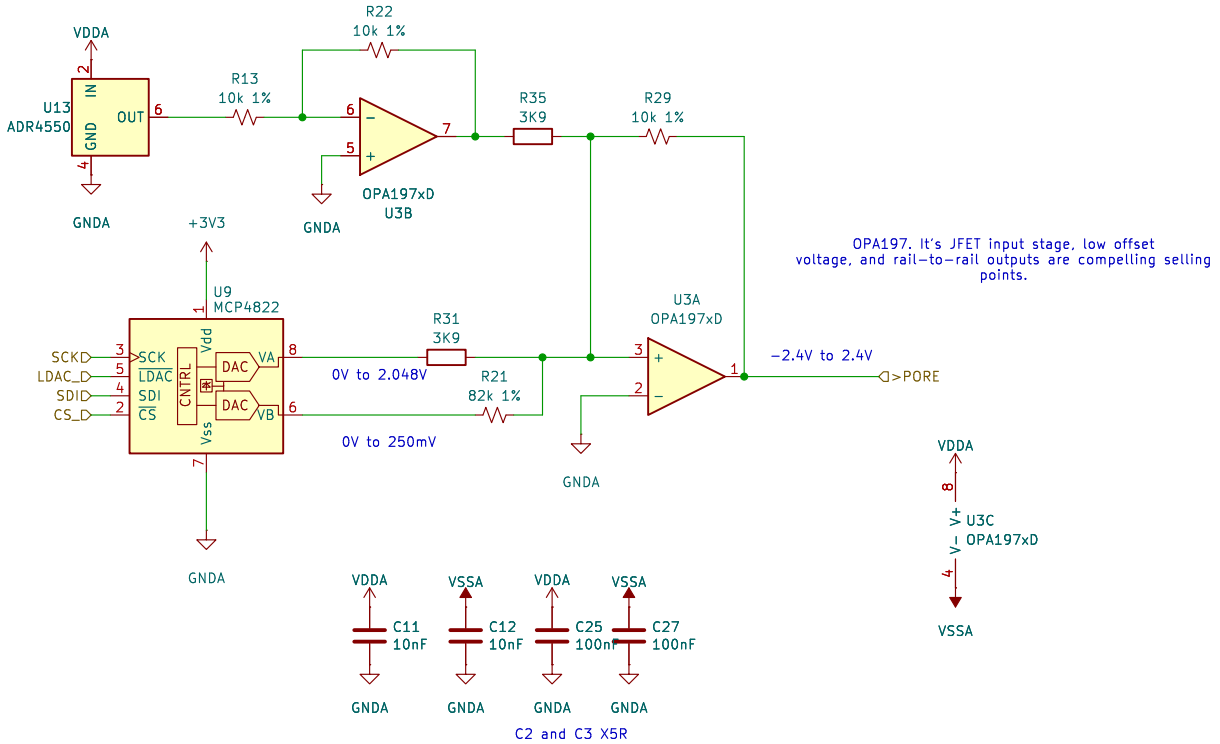
250ft 32AWG CAT7 STP Solid, 0.075" Super Flat Bulk Ethernet Bare Copper Cable – White.

SSH-003T-P0.2-H #32 to #28 (0.032 to 0.081) 0.4 to 0.8 23.000

Sheet: /Insturmentation/		
File: Insturmentation.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.6		Id: 4/5

use a precision reference here

−1.024V



Sheet: /Ignition/		
File: Ignition.kicad_sch		
Title:		
Size: A4	Date:	**Rev:**
KiCad E.D.A. 9.0.6		Id: 5/5